# Gleipnir: Frontier-Quality Control Monitors at a Fraction of the Cost

## Sprint proposal: from building the monitors to stress-testing them

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# Motivation and core idea

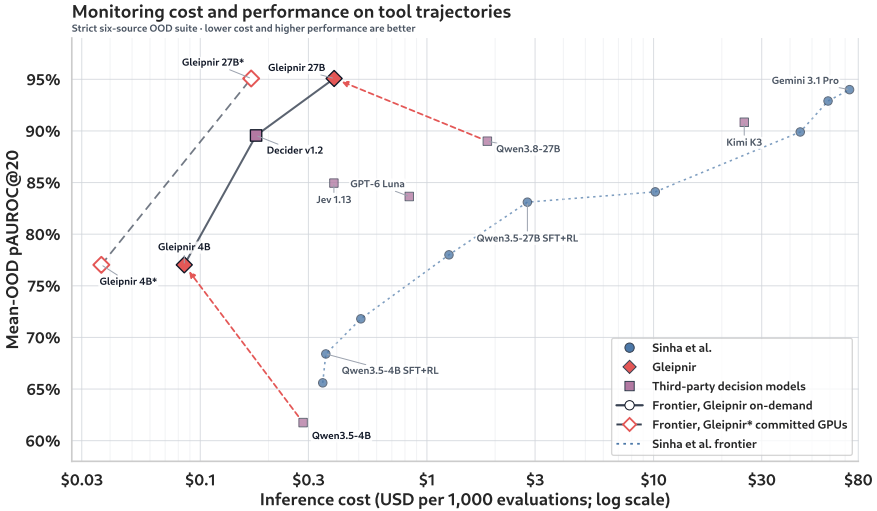
## Motivation

Make monitoring cheap enough to run on every agent action everywhere. Global deployment makes misalignment much easier to detect.

- **Prior work** distills a teacher's written rationale and score, so each evaluation generates hundreds of tokens.
- **Gleipnir** asks the teacher (Kimi K3) for a single decision, `0` or `1`, and distills its **probability** on that token. Each evaluation is one forward pass over the trajectory.

Sinha et al. (2026), *Training Deliberative Monitors for Black-Box Scheming Detection*.

# Frontier quality at about 1/200 of the cost




6,395 OOD trajectories from six sources (Sinha et al.). Gleipnir at self-hosted B200 break-even; * committed GPUs.

# Status: ahead of the proposal

**Proposal, week 1: improve the monitors**

- Done. Gleipnir 4B (seed average + quantization-aware distillation) and Gleipnir 27B are frozen release candidates.
- Optimized B200 serving and release packaging are built.

**Proposal, week 2: stress-test them**

- Starting now: the whole sprint goes to the second half of the proposal.
- Both monitors are frozen, so every result describes the release checkpoints.

## Sprint deliverable

A **model card** for each monitor that tells the community what it is good at, where it fails and how easily it can be fooled.

# Four avenues for the model card

## 1. Fresh benchmarks

One-shot evaluation of the frozen monitors on benchmarks never used in the project (MonitoringBench, SleightBench).

*Low effort*

## 2. Black-box vs. white-box monitors

Compare against probes. We know of no benchmark with *open-model* agents on control tasks; building one is slow.

*High effort*

## 3. Attacks and mechanistic fixes

The weights are public, so white-box attacks on one decision logit are easy to find. How easily does a decision flip? Can ablating directions remove unwanted traits?

*Medium effort*

## 4. Newly affordable domains

Settings that were too costly to monitor before, such as agent swarms: one million actions cost \$390 with Gleipnir 27B compared with \$73K with Gemini 3.1 Pro.

*Medium effort*

# Where we would like feedback

- **Priorities.** Which of the four avenues would make the model card most useful to the control community in two weeks?
- **Benchmarks.** Beyond MonitoringBench and SleightBench, which results would be most convincing?
- **White-box comparison.** Is there existing data with open-model agents on control tasks, or is building our own worth the time?
- **Attacks.** Which attack settings would be credible: white-box only, or black-box query attacks as for an API-served monitor too?
- **New domains.** Which setting would cheap monitoring unlock that matters most: agent swarms or long-horizon sessions?

# Logistics

- **TA.** Could we be paired with a TA who is interested in AI control and monitoring?
- **Schedule.** Jeremias attends EMNLP from 23 to 30 October, so his productivity will be reduced that week. Could we extend the sprint by working on 31 October and 1 November?